

MEER10

A 64-bit hash finalizer that goes the full diploma distance for ten points.

256 streams of one terabyte each, no failure. The cheapest chain known to us that survives RRC-64-40. This datasheet documents the whole cascade, the two places where the mixer is weaker than the one it succeeds, and everything that was not measured.

Instruments that do not question themselves lie.

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The construction

```
uint64_t meer10(uint64_t x) {
    x = mulfold(x, 0x5846DFED2F0E1D49);
    x ^= (x >> 28) ^ (x >> 6);
    return mulfold(x, 0x781F94B96E8EDB3B);
}
```

mulfold(v, c) is the 128-bit fold: multiply v by c to the full 128 bits, then XOR the two 64-bit halves together. Two folds with a double xor-shift between them — no constant on the input. That last detail separates this chain from every other passer in this work, and it has its own section below.

Property	Value
Cost (latency model of this work)	10 = 4 + 2 + 4 (fold, double xor-shift, fold)
Class	hash finalizer — collisions allowed, not bijective
State	64 bit, no side effect, no memory
Avalanche (measured, 8192 inputs)	0.5002
Collisions	0 across 2^{20} consecutive inputs
meer10(0)	0 — a fixed point at zero, see section below
Portability	anywhere $64 \times 64 \rightarrow 128$ -bit is available

Both constants are borrowed rather than derived: 0x781F94... is the fold constant from our own earlier search that also carries lu9, and 0x5846DF... comes from the same pool. Composing them into this chain, and measuring the chain to diploma distance, is what this paper contributes; the skeleton is not.

Origin: found, not designed

This is where this paper differs from the one on lu9, and the difference is not flattering. lu9 was *constructed*: a measured weakness was named, a repair was reasoned out, and the repair worked. meer10 was *found*. It came out of a search over an extended operation alphabet, at generation 10, and its record in the archive reads:

```
mf0x5846dfed2f0e1d49 xsh(28,6) mf0x781f94b96e8edb3b
cost 10    gauntlet 4/4    smoke 32/32
```

No hypothesis preceded it. The search proposed the chain, the ladder let it through, and only afterwards was there a reason to look at why. Anyone who prefers their results to come with a derivation should discount this one accordingly. The result is a measurement, not an insight into how to build such a chain from scratch.

What the search is, and what stays private.

The search itself is not part of this publication and will not be. What is published is everything it produced that can be checked independently: the chain, its constants, the byte streams it generates, and every verdict the ladder handed down. None of that requires trusting the search — the chain is six lines of C, and anyone can push it through the same rig.

Two forms, one stream.

meer10 runs under its name or as a generic chain:

```
feeder meer10 0 0 0
feeder chain 0 0 0 3 8 5846dfed2f0e1d49 \
10 706 8 781f94b96e8edb3b
```

The two forms were checked against each other rather than assumed equal: 4 MB of each, MD5 8408aad25ff37597912bcd4247e334fb, identical. The diploma was run under the name; the chain form exists so that nobody has to recompile anything to reproduce it.

The full measurement protocol

Measured on the RRC ladder: counting numbers in disguises (read backwards, complemented, rotated), judged by PractRand. Every stage carried anchors. A known weak mixer had to fall in the same run, and from 28 August a strong-side anchor had to hold as well. One qualification concerns this very run: until 29 August that upper slot was filled by our own mfx9, which is a pre-flight check on the candidate and not a calibration weight, because nobody outside this rig had ever measured it. The diploma reported here therefore ran with one published weight, mix13 falling, and not two. THE CALIBRATION WEIGHTS, section 8.1, sets out why that distinction matters.

Stage	Scope	Result
Gauntlet	4 streams at 64 MB	holds, 4/4
Smoke	32 streams at 2 GB	holds, 32/32
Depth test	64 streams at 16 GB (2^{34})	holds, 64/64 — 0 FAIL, 0 aborted, 0 short
Diploma RRC-64-40	256 streams at 1 TB (2^{40})	HOLDS, 256/256 — not one FAIL in 256 terabytes

The depth stage was run last, on 28 August, and out of order.

It belongs between smoke and diploma, and meer10 went straight from the 2 GB smoke into the 1 TB diploma while several papers stated the depth test as passed. The diploma subsumes that stage in every dimension, so the claim was almost certainly true — but it was a claim about a run that had not happened, so the run was performed: 39 minutes, 64 of 64 clean, evidence in `results/depth_meer10_result.json`. The same gap existed for lu9 and was closed the same way on 30 August, also 64 of 64.

What the diploma cost.

256 terabytes of generated and judged stream, ten streams at a time, in at least three run segments between 22 and 28 August 2026. What that came to in total is not recorded: the result file carries 39.5 hours, and that is the last segment only, because the clock starts with the process while the stream count comes from the checkpoint file. The correction is dated 31 August 2026. PractRand reported **34 suspicious results** in total, none of them a failure: 224 of the 256 streams came through with nothing at all to note, 30 with one, and 2 with two. A suspicious result is PractRand's weakest flag, one rung below a finding; at 256 terabytes across roughly a quarter of a million sub-tests, a scattering of them is expected.

The evidence, and the day it nearly was not

Four of the 256 raw logs were destroyed by this rig's own anchor. Before every start the diploma runner re-checks its candidate on four gauntlet rotations — and those four rotations are also four of the 256 diploma streams. Same mixer, same rotations, same filename, opened in truncating mode. Every restart overwrote the 1 TB logs of those four streams with a 256 MB anchor result, and resume then skipped them because the checkpoint had booked them already.

The verdicts were real: the function that produces them reads nothing but file content, and it recorded “clean to 2^{40} ” at a time when the files still said so. Their booked elapsed times place them within seconds of six streams whose complete 1 TB logs survive. The reconstruction held up. It was still only a reconstruction, so the four streams were re-measured from scratch on 28 August, after the overwrite was fixed: all four came back clean to 2^{40} , identical to what the checkpoint had recorded seven days earlier.

All 256 raw logs are complete on disk today. The incident is treated in *The Calibration Weights*, where it belongs; it is named here because the evidence came close to being unusable, and that belongs in the datasheet.

Why it holds where nine tears

meer10 carries the same skeleton as the specious family (fold, something, fold) and the same skeleton as Maiga's mxm, with 128-bit folds in place of plain multiplications. All three repair in the middle rather than at the input. Two of them die; one goes the distance.

Chain	Middle step	Cost	Ladder
specious family	bswap — rearranges bytes	9	dies 2^{31} – 2^{34}
lu9	— (constant at the <i>input</i>)	9	tears at 2^{38}
meer10	$x \wedge= (x \gg 28) \wedge (x \gg 6)$ — diffuses	10	holds 2^{40}

A byte swap moves bits; a double xor-shift mixes them. Same position in the chain, and the one hides the weakness while the other heals it. That is also the mechanical answer to a question the cost-limit paper left standing: how little separates the ladder's entry test from its diploma. At nine, the middle member can only afford a permutation; the next point up pays for real mixing. (What that point is worth depends on a weight the repository README argues with itself about: one point under the table as printed, two if the double shift-xor is priced like its rotate twin.)

Why the low bits survive.

lu9 tears at 2^{38} in the lowest bits, and the reason is structural: in a multiplication, carries only travel upwards, so output bit k depends only on input bits 0 to k . The fold repairs that partially. meer10's middle step shifts right by 28 and by 6 and XORs both back in — information moves *downward*, into exactly the bits that carry the residual structure on counter streams. That is a mechanism consistent with the measurement; it is not a proof that this is why the chain holds.

The fixed point at zero

meer10(0) = 0. The chain has no constant on the input, so the first fold maps zero to zero, the xor-shift leaves it there, and the second fold maps it to zero again. This is not inferred: the first eight bytes of the stream T0C0r00, taken from the shipped feeder binary, are zero.

00000000: 0000 0000 0000 0000 6258 e32c 005f 4f1b

The whole of *LU9* turns on that same property. The bare double fold maps zero to zero with every constant, *lu9* was built to remove exactly this by putting a constant in front, and the removal worked — *lu9(0)* is 0x37F28927366A4E7B. *meer10* does not remove it. It leaves the fixed point standing and passes the diploma anyway, on streams that begin at zero and count up, where *lu9* tears.

What follows from that, and what does not.

This is a correction to the mechanism story: **the zero fixed point is not what the RRC ladder punishes.** The ladder punishes the statistical thinness of products from low-weight inputs, and a diffusing middle step fixes that without touching the fixed point at all. The earlier papers treated two separate defects as one.

What does not follow is that the fixed point is harmless. It is a real property with real consequences outside this rig: hash a record whose identifier is zero and you get zero back; seed a counter-based generator with zero and the first output is zero. Anyone for whom that matters should put a constant on the input (one operation, cost +1) and then re-measure, because the resulting chain is not the one this paper documents.

This is a place where meer10 is worse than the mixer it outperforms on the ladder.

The reach — what meer10 can do

On structured counter streams, meer10 shows no detectable weakness up to 2^{40} bytes per disguise, across 256 disguises. No stream failed.

That is the deepest measurement in this work and, at cost 10, the cheapest chain we know of that reaches it. For comparison: lu9 (cost 9) tears at 2^{38} , and the two published mixers known to pass RRC at this depth or beyond cost around fifteen and seventeen points in this model.

Mixer	Cost	Deepest measurement
mix13 (Stafford)	12	dies 2^{16} – 2^{21}
fmix64 (Appleby)	12	dies 2^{14} – 2^{18}
Moremur (Evensen)	12	does not pass RRC-64-40
lu9	9	tears at 2^{38}
meer10	10	passes RRC-64-40 (2^{40}), 256/256
NASAM (Evensen)	~15	passes RRC-64-42
mx3 rev2 (Kågstöm)	~17	passes RRC-64-42

What that table does not say.

It does not say meer10 is better than NASAM or mx3. Those two pass a stage *deeper* than anything measured here: RRC-64-42, four times the data per stream. meer10 has never been run at 2^{42} and might not survive it. The defensible claim is narrow: **at the RRC-64-40 distance, meer10 is the cheapest passer known to us.** Whether it holds deeper is unmeasured, and whether something cheaper passes is unknown, because nobody can enumerate the 64-bit constant space.

For ordinary use this is far past sufficient. A hash table, a sharding key or a fingerprint never sees a terabyte of mixed output from a single counter sequence. The measurement matters for long generator streams and large simulations — and there, the choice between meer10 and NASAM is a choice between five cost points and one further stage of proven depth.

What was NOT measured

This paper does **not** answer the following:

RRC-64-42 and beyond.

The stage at which NASAM and mx3 rev2 are documented has not been attempted. It is four times the data per stream and roughly six days of further measurement. Until it is run, meer10's depth is 2^{40} and nothing more.

Speed.

The cost figure 10 is a weight table, and on 31 August 2026 it was measured against a clock for the first time. A dependent chain on a Ryzen 7 5700X puts meer10 at **3.28 ns per call**: faster than NASAM at 3.84, and **slower than mix13, fmix64 and moremur at 3.01 to 3.07**, which the weight table prices two points higher. So the table ranks meer10 below them and the clock does not. The saving this paper claims is real against NASAM and not against the classical finalizers. 100 million iterations, three runs, spread under 0.17 ns, `results/cost_bench.json`; throughput and other CPUs remain unmeasured.

Collision behaviour at scale.

2^{20} consecutive inputs were checked (zero collisions). A sample of that size is not evidence of collision quality. As a fold, meer10 is not bijective; collisions necessarily exist.

Behaviour on real keys.

The measurement covers structured counter streams — the hardest realistic case for a finalizer. Strings, records and complete hash functions are answered by other tools (SMHasher and relatives).

Security.

meer10 is **not** cryptographic. It has no one-way property: with the constants known, preimages are cheap to find. (As a fold it is not a bijection, so it cannot be inverted as a function; that is no protection, because an attacker needs only *some* input reaching a given output.) It has a fixed point at zero, and it offers no protection against deliberately constructed inputs.

Novelty.

A literature check on 20 August found this combination nowhere. One search turning up nothing is a negative result, and no proof of novelty. Nor is it a claim of priority: the skeleton is Maiga's, the fold as a mixing step goes back to Makarov and the wyhash line, and the constants are borrowed. New here are the chain and the measurement of it.

And the limit of the statement itself.

That meer10 holds 2^{40} on 256 disguises is measured. That it has no weakness below that is *not proven*. It has not been demonstrated, which is a different thing. A sharper tool could find something the ladder does not.

Measuring it yourself

The rig is documented openly (feeder, driver, PractRand with build instructions). The verdict depends only on the byte stream, not on the machine, so no access to ours is required. The diploma is roughly two days of compute; the depth stage is 39 minutes and settles the question for most purposes.

Provenance of the building blocks

Skeleton (mix, shift, mix): Jon Maiga's mxm, here with 128-bit folds instead of plain multiplications.

Fold as a mixing step: Vladimir Makarov (mum-hash), carried on in the wyhash line by Wang Yi.

Why the lowest bits are weak: Klimov and Shamir's T-functions (2004). **Test scheme**: Pelle Evensen's RRC. **Test program**: PractRand, Chris Doty-Humphrey. **Plotting cost against RRC depth**: Martin Leitner-Ankerl (2020). **The chain, the constants, the cascade out to diploma distance and the name**: this work.

In brief

What it is	A 64-bit hash finalizer: two 128-bit folds with a double xor-shift between them, cost 10.
Where it comes from	Found by a search, not derived. No hypothesis preceded it.
What it holds	4/4 gauntlet, 32/32 smoke, 64/64 depth (2^{34}), and 256/256 streams at 1 TB — the diploma, RRC-64-40, without a single failure.
What it costs	Ten points, against roughly fifteen and seventeen for the published mixers known to pass RRC at this depth or deeper.
Where it is weak	$\text{meer10}(0) = 0$. The fixed point the K8 weakness is famous for is still there — the ladder simply does not punish it. Not cryptographic, not bijective, untested beyond 2^{40} .
The honest claim	At the RRC-64-40 distance, the cheapest passer known to us. Not the best mixer; NASAM and mx3 rev2 are documented one stage deeper.

Instruments that do not question themselves lie.

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As of: 28 August 2026. Measuring rig: *The Ladder*. Cost bounds: *The Cost Limit*. The chain that tears one point cheaper: *LU9* and *Where K9 Ends*. The same skeleton, one point cheaper and dead: *The Specious Family*. The anchor incident: *The Calibration Weights*.